### MySQL Lab-1

### Question 1: Student Management System

### Table Structure

| Column | Data Type |
|---|---|
| student_id | INT (Primary Key) |
| name | VARCHAR(50) |
| age | INT |
| city | VARCHAR(30) |
| course | VARCHAR(30) |
| marks | INT |

### Sample Data (20 Records)

| ID | Name | Age | City | Course | Marks |
|---|---|---|---|---|---|
| 101 | Amit | 20 | Delhi | BCA | 78 |
| 102 | Priya | 21 | Pune | BSc | 89 |
| 103 | Rahul | 22 | Mumbai | BCA | 67 |
| 104 | Sneha | 20 | Delhi | BTech | 92 |
| 105 | Rohan | 23 | Jaipur | BSc | 75 |
| 106 | Anjali | 21 | Pune | BCA | 81 |
| 107 | Vikas | 22 | Mumbai | BTech | 64 |
| 108 | Neha | 20 | Delhi | BCA | 90 |
| 109 | Arjun | 23 | Lucknow | BSc | 55 |
| 110 | Kiran | 21 | Pune | BCA | 73 |
| 111 | Meera | 22 | Delhi | BTech | 95 |
| 112 | Nikhil | 20 | Mumbai | BSc | 70 |
| 113 | Pooja | 21 | Jaipur | BCA | 84 |
| 114 | Ritesh | 22 | Pune | BTech | 60 |
| 115 | Komal | 20 | Delhi | BCA | 88 |
| 116 | Deepak | 23 | Mumbai | BSc | 77 |
| 117 | Sonal | 21 | Pune | BCA | 69 |
| 118 | Manish | 22 | Delhi | BTech | 82 |
| 119 | Ankit | 20 | Lucknow | BCA | 91 |
| 120 | Payal | 21 | Jaipur | BSc | 72 |

### Practice Tasks

1. Create above table in database.

```
mysql> create database Student_Management_System_db;
Query OK, 1 row affected (0.01 sec)

mysql> use Student_Manangement_System_db;
ERROR 1049 (42000): Unknown database 'student_manangement_system_db'
mysql> use Student_Management_System_db;
Database changed
mysql> create table Students
    -> cret;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right
 syntax to use near 'cret' at line 2
mysql> create table Students( student_id int primary key,name varchar(50),age int ,city varchar(30),course varchar(30),marks int);
Query OK, 0 rows affected (0.03 sec)

mysql> desc Students;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| student_id | int         | NO   | PRI | NULL    |       |
| name       | varchar(50) | YES  |     | NULL    |       |
| age        | int         | YES  |     | NULL    |       |
| city       | varchar(30) | YES  |     | NULL    |       |
| course     | varchar(30) | YES  |     | NULL    |       |
| marks      | int         | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

2. Insert all 20 records.

```
mysql> insert into Students values(101,'Amit',20,'Delhi','BCA',78);
Query OK, 1 row affected (0.01 sec)

mysql> insert into Students values(102,'Priya',21,'Pune','BSc',89);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(103,'Rahul',22,'Mumbai','BCA',67);
Query OK, 1 row affected (0.00 sec)

mysql> ^C
mysql> insert into Students values(104,'Sneha',20,'Delhi','BTech',92);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(105,'Rohan',23,'Jaipur','BSc',75);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(106,'Anjali',21,'Pune','BCA',81);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(107,'Vikas',22,'Mumbai','BTech',64);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(108,'Arjun',23,'Lucknow','BSc',55);
Query OK, 1 row affected (0.01 sec)

mysql> insert into Students values(109,'Neha',20,'Delhi','BCA',90);
Query OK, 1 row affected (0.01 sec)

mysql> insert into Students values(110,'Kiran',21,'Pune','BCA',73);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(111,'Meera',22,'Delhi','BTech',95);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(112,'Nikhil',20,'Mumbai','BSc',70);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(113,'Pooja',21,'Jaipur','BCA',84);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(114,'Ritesh',22,'Pune','BTech',60);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(115,'Komal',20,'Delhi','BCA',78);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(116,'Deepak',23,'Mumbai','BSc',77);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(117,'Sonal',21,'Pune','BCA',69);
Query OK, 1 row affected (0.01 sec)

mysql> insert into Students values(118,'Manish',22,'Delhi','BTech',82);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(119,'Ankit',20,'Lucknow','BCA',91);
Query OK, 1 row affected (0.00 sec)

mysql> insert into Students values(120,'Payal',21,'Jaipur','BSc',72);
Query OK, 1 row affected (0.00 sec)
```

3. Display all student records.

```
mysql> select * from Students;
+------------+---------+------+---------+--------+-------+
| student_id | name    | age  | city    | course | marks |
+------------+---------+------+---------+--------+-------+
|        101 | Amit    |   20 | Delhi   | BCA    |    78 |
|        102 | Priya   |   21 | Pune    | BSc    |    89 |
|        103 | Rahul   |   22 | Mumbai  | BCA    |    67 |
|        104 | Sneha   |   20 | Delhi   | BTech  |    92 |
|        105 | Rohan   |   23 | Jaipur  | BSc    |    75 |
|        106 | Anjali  |   21 | Pune    | BCA    |    81 |
|        107 | Vikas   |   22 | Mumbai  | BTech  |    64 |
|        108 | Arjun   |   23 | Lucknow | BSc    |    55 |
|        109 | Neha    |   20 | Delhi   | BCA    |    90 |
|        110 | Kiran   |   21 | Pune    | BCA    |    73 |
|        111 | Meera   |   22 | Delhi   | BTech  |    95 |
|        112 | Nikhil  |   20 | Mumbai  | BSc    |    70 |
|        113 | Pooja   |   21 | Jaipur  | BCA    |    84 |
|        114 | Ritesh  |   22 | Pune    | BTech  |    60 |
|        115 | Komal   |   20 | Delhi   | BCA    |    78 |
|        116 | Deepak  |   23 | Mumbai  | BSc    |    77 |
|        117 | Sonal   |   21 | Pune    | BCA    |    69 |
|        118 | Manish  |   22 | Delhi   | BTech  |    82 |
|        119 | Ankit   |   20 | Lucknow | BCA    |    91 |
|        120 | Payal   |   21 | Jaipur  | BSc    |    72 |
+------------+---------+------+---------+--------+-------+
20 rows in set (0.00 sec)
```

4. Display only the names and marks of all students.

```
mysql> select name,marks from Students;
+---------+-------+
| name    | marks |
+---------+-------+
| Amit    |    78 |
| Priya   |    89 |
| Rahul   |    67 |
| Sneha   |    92 |
| Rohan   |    75 |
| Anjali  |    81 |
| Vikas   |    64 |
| Arjun   |    55 |
| Neha    |    90 |
| Kiran   |    73 |
| Meera   |    95 |
| Nikhil  |    70 |
| Pooja   |    84 |
| Ritesh  |    60 |
| Komal   |    78 |
| Deepak  |    77 |
| Sonal   |    69 |
| Manish  |    82 |
| Ankit   |    91 |
| Payal   |    72 |
+---------+-------+
20 rows in set (0.00 sec)
```

5. Find students who belong to Delhi.

```
mysql>  select * from Students where city = 'Delhi';
+------------+--------+------+-------+--------+-------+
| student_id | name   | age  | city  | course | marks |
+------------+--------+------+-------+--------+-------+
|        101 | Amit   |   20 | Delhi | BCA    |    78 |
|        104 | Sneha  |   20 | Delhi | BTech  |    92 |
|        109 | Neha   |   20 | Delhi | BCA    |    90 |
|        111 | Meera  |   22 | Delhi | BTech  |    95 |
|        115 | Komal  |   20 | Delhi | BCA    |    78 |
|        118 | Manish |   22 | Delhi | BTech  |    82 |
+------------+--------+------+-------+--------+-------+
6 rows in set (0.00 sec)
```

6.Display students whose marks are greater than 80.

```
mysql> select * from Students where course = 'BCA';
+------------+---------+------+---------+--------+-------+
| student_id | name    | age  | city    | course | marks |
+------------+---------+------+---------+--------+-------+
|        101 | Amit    |   20 | Delhi   | BCA    |    78 |
|        103 | Rahul   |   22 | Mumbai  | BCA    |    67 |
|        106 | Anjali  |   21 | Pune    | BCA    |    81 |
|        109 | Neha    |   20 | Delhi   | BCA    |    90 |
|        110 | Kiran   |   21 | Pune    | BCA    |    73 |
|        113 | Pooja   |   21 | Jaipur  | BCA    |    84 |
|        115 | Komal   |   20 | Delhi   | BCA    |    78 |
|        117 | Sonal   |   21 | Pune    | BCA    |    69 |
|        119 | Ankit   |   20 | Lucknow | BCA    |    91 |
+------------+---------+------+---------+--------+-------+
9 rows in set (0.00 sec)
```

7.Find all BCA students.

```
mysql> select * from Students where course = 'BCA';
+------------+---------+------+---------+--------+-------+
| student_id | name    | age  | city    | course | marks |
+------------+---------+------+---------+--------+-------+
|        101 | Amit    |   20 | Delhi   | BCA    |    78 |
|        103 | Rahul   |   22 | Mumbai  | BCA    |    67 |
|        106 | Anjali  |   21 | Pune    | BCA    |    81 |
|        109 | Neha    |   20 | Delhi   | BCA    |    90 |
|        110 | Kiran   |   21 | Pune    | BCA    |    73 |
|        113 | Pooja   |   21 | Jaipur  | BCA    |    84 |
|        115 | Komal   |   20 | Delhi   | BCA    |    78 |
|        117 | Sonal   |   21 | Pune    | BCA    |    69 |
|        119 | Ankit   |   20 | Lucknow | BCA    |    91 |
+------------+---------+------+---------+--------+-------+
9 rows in set (0.00 sec)
```

**Question 2: Employee Database**

**Table Structure**

| **Column** | **Data Type** |
|---|---|
| emp_id | INT (Primary Key) |
| emp_name | VARCHAR(50) |
| department | VARCHAR(30) |
| salary | INT |
| city | VARCHAR(30) |
| experience | INT |

**Sample Data (20 Records)**

| **ID** | **Name** | **Department** | **Salary** | **City** | **Experience** |
|---|---|---|---|---|---|
| 1 | Amit | HR | 35000 | Delhi | 2 |
| 2 | Priya | IT | 60000 | Pune | 5 |
| 3 | Rahul | Sales | 45000 | Mumbai | 4 |
| 4 | Sneha | IT | 75000 | Delhi | 7 |
| 5 | Rohan | Finance | 50000 | Jaipur | 3 |

| | | | | | |
|---|---|---|---|---|---|
| 6 | Anjali | HR | 42000 | Pune | 2 |
| 7 | Vikas | IT | 82000 | Mumbai | 8 |
| 8 | Neha | Sales | 39000 | Delhi | 2 |
| 9 | Arjun | Finance | 68000 | Lucknow | 6 |
| 10 | Kiran | IT | 56000 | Pune | 4 |
| 11 | Meera | HR | 47000 | Delhi | 3 |
| 12 | Nikhil | Sales | 51000 | Mumbai | 5 |
| 13 | Pooja | Finance | 72000 | Jaipur | 7 |
| 14 | Ritesh | IT | 61000 | Pune | 5 |
| 15 | Komal | HR | 38000 | Delhi | 2 |
| 16 | Deepak | Finance | 65000 | Mumbai | 6 |
| 17 | Sonal | IT | 78000 | Pune | 8 |
| 18 | Manish | Sales | 44000 | Delhi | 3 |
| 19 | Ankit | IT | 59000 | Lucknow | 4 |
| 20 | Payal | Finance | 71000 | Jaipur | 7 |

**Practice Tasks**

1. Create Above table in database.

```
mysql> desc Employee;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| emp_id     | int         | NO   | PRI | NULL    |       |
| emp_name   | varchar(50) | YES  |     | NULL    |       |
| department | varchar(30) | YES  |     | NULL    |       |
| salary     | int         | YES  |     | NULL    |       |
| city       | varchar(30) | YES  |     | NULL    |       |
| experience | int         | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

2. Insert above 20 records.
3. Display all employee details.

```
mysql> select * from Employee;
+--------+----------+------------+--------+---------+------------+
| emp_id | emp_name | department | salary | city    | experience |
+--------+----------+------------+--------+---------+------------+
|      1 | Amit     | HR         |   3500 | Delhi   |          2 |
|      2 | Priya    | IT         |  60000 | Pune    |          5 |
|      3 | Rahul    | Sales      |  45000 | Mumbai  |          4 |
|      4 | Sneha    | Finance    |  50000 | Jaipur  |          7 |
|      5 | Rohan    | IT         |  75000 | Delhi   |          2 |
|      6 | Anjali   | HR         |  42000 | Pune    |          7 |
|      7 | Vikas    | IT         |  82000 | Mumbai  |          8 |
|      8 | Neha     | Sales      |  39000 | Delhi   |          2 |
|      9 | Arjun    | Finance    |  68000 | Lucknow |          6 |
|     10 | Kiran    | IT         |  56000 | MumPune |          4 |
|     11 | Meera    | HR         |  47000 | Delhi   |          3 |
|     12 | Nikhil   | Sales      |  51000 | Mumbai  |          5 |
|     13 | Pooja    | Finance    |  72000 | Jaipur  |          7 |
|     14 | Ritesh   | IT         |  61000 | Pune    |          5 |
|     15 | Komal    | HR         |  38000 | Delhi   |          2 |
|     16 | Deepak   | Finance    |  65000 | Mumbai  |          6 |
|     17 | Sonal    | IT         |  78000 | Pune    |          8 |
|     18 | Manish   | Sales      |  44000 | Delhi   |          3 |
|     19 | Ankit    | IT         |  59000 | Lucknow |          4 |
|     20 | Payal    | Finance    |  71000 | Lucknow |          7 |
+--------+----------+------------+--------+---------+------------+
20 rows in set (0.00 sec)
```

4. Display employee names and salaries.

```
mysql> select emp_name, salary from Employee;
+----------+--------+
| emp_name | salary |
+----------+--------+
| Amit     |   3500 |
| Priya    |  60000 |
| Rahul    |  45000 |
| Sneha    |  50000 |
| Rohan    |  75000 |
| Anjali   |  42000 |
| Vikas    |  82000 |
| Neha     |  39000 |
| Arjun    |  68000 |
| Kiran    |  56000 |
| Meera    |  47000 |
| Nikhil   |  51000 |
| Pooja    |  72000 |
| Ritesh   |  61000 |
| Komal    |  38000 |
| Deepak   |  65000 |
| Sonal    |  78000 |
| Manish   |  44000 |
| Ankit    |  59000 |
| Payal    |  71000 |
+----------+--------+
20 rows in set (0.00 sec)
```

5. Find employees whose salary is greater than ₹60,000

```
mysql> select *from  Employee where salary > 60000;
+--------+----------+------------+--------+---------+------------+
| emp_id | emp_name | department | salary | city    | experience |
+--------+----------+------------+--------+---------+------------+
|      5 | Rohan    | IT         |  75000 | Delhi   |          2 |
|      7 | Vikas    | IT         |  82000 | Mumbai  |          8 |
|      9 | Arjun    | Finance    |  68000 | Lucknow |          6 |
|     13 | Pooja    | Finance    |  72000 | Jaipur  |          7 |
|     14 | Ritesh   | IT         |  61000 | Pune    |          5 |
|     16 | Deepak   | Finance    |  65000 | Mumbai  |          6 |
|     17 | Sonal    | IT         |  78000 | Pune    |          8 |
|     20 | Payal    | Finance    |  71000 | Lucknow |          7 |
+--------+----------+------------+--------+---------+------------+
8 rows in set (0.00 sec)
```

6. Find employees working in the IT department.

```
mysql> select * from Employee where department = 'IT';
+--------+----------+------------+--------+---------+------------+
| emp_id | emp_name | department | salary | city    | experience |
+--------+----------+------------+--------+---------+------------+
|      2 | Priya    | IT         |  60000 | Pune    |          5 |
|      5 | Rohan    | IT         |  75000 | Delhi   |          2 |
|      7 | Vikas    | IT         |  82000 | Mumbai  |          8 |
|     10 | Kiran    | IT         |  56000 | MumPune |          4 |
|     14 | Ritesh   | IT         |  61000 | Pune    |          5 |
|     17 | Sonal    | IT         |  78000 | Pune    |          8 |
|     19 | Ankit    | IT         |  59000 | Lucknow |          4 |
+--------+----------+------------+--------+---------+------------+
7 rows in set (0.00 sec)
```

7. Display employees having more than 5 years of experience

```
mysql> select * from Employee where experience > 5;
+--------+----------+------------+--------+---------+------------+
| emp_id | emp_name | department | salary | city    | experience |
+--------+----------+------------+--------+---------+------------+
|      4 | Sneha    | Finance    |  50000 | Jaipur  |          7 |
|      6 | Anjali   | HR         |  42000 | Pune    |          7 |
|      7 | Vikas    | IT         |  82000 | Mumbai  |          8 |
|      9 | Arjun    | Finance    |  68000 | Lucknow |          6 |
|     13 | Pooja    | Finance    |  72000 | Jaipur  |          7 |
|     16 | Deepak   | Finance    |  65000 | Mumbai  |          6 |
|     17 | Sonal    | IT         |  78000 | Pune    |          8 |
|     20 | Payal    | Finance    |  71000 | Lucknow |          7 |
+--------+----------+------------+--------+---------+------------+
8 rows in set (0.00 sec)
```

**Question 3: Product Inventory**

**Table Structure**

| Column | Data Type |
| --- | --- |
| product_id | INT (Primary Key) |
| product_name | VARCHAR(50) |
| category | VARCHAR(30) |
| price | INT |
| quantity | INT |
| supplier | VARCHAR(40) |

**Sample Data (20 Records)**

| Product ID | Product Name | Category | Price (₹) | Quantity | Supplier |
| --- | --- | --- | --- | --- | --- |
| 101 | Laptop | Electronics | 55000 | 15 | Dell |
| 102 | Mouse | Electronics | 799 | 120 | Logitech |
| 103 | Keyboard | Electronics | 1499 | 80 | HP |
| 104 | Monitor | Electronics | 12500 | 30 | Samsung |
| 105 | Printer | Electronics | 8500 | 20 | Canon |
| 106 | Office Chair | Furniture | 4500 | 25 | Godrej |
| 107 | Study Table | Furniture | 6500 | 18 | Nilkamal |
| 108 | Bookshelf | Furniture | 7200 | 12 | Godrej |
| 109 | Sofa | Furniture | 28000 | 8 | Urban Ladder |
| 110 | Dining Table | Furniture | 22000 | 10 | IKEA |
| 111 | Water Bottle | Kitchen | 450 | 150 | Milton |
| 112 | Mixer Grinder | Kitchen | 3800 | 22 | Prestige |
| 113 | Pressure Cooker | Kitchen | 2800 | 35 | Hawkins |
| 114 | Gas Stove | Kitchen | 5200 | 16 | Sunflame |
| 115 | Microwave Oven | Kitchen | 9800 | 14 | LG |
| 116 | Refrigerator | Electronics | 36000 | 9 | Whirlpool |
| 117 | Ceiling Fan | Electronics | 2500 | 40 | Crompton |
| 118 | Air Conditioner | Electronics | 42000 | 7 | Voltas |
| 119 | Smart TV | Electronics | 48000 | 11 | Sony |
| 120 | Iron Box | Electronics | 1800 | 55 | Philips |

**Practice Tasks**

1. Create above table.

```
mysql> create database Product_Inventory_db;
Query OK, 1 row affected (0.01 sec)

mysql> use Product_Inventory_db;
Database changed
```

```
mysql> desc Product;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| product_id   | int         | NO   | PRI | NULL    |       |
| product_name | varchar(50) | YES  |     | NULL    |       |
| category     | varchar(30) | YES  |     | NULL    |       |
| price        | int         | YES  |     | NULL    |       |
| quantity     | int         | YES  |     | NULL    |       |
| supplier     | varchar(40) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
6 rows in set (0.01 sec)
```

2. Insert above 20 records.
3. Display all products.

```
mysql> select * from Product;
+------------+------------------+-------------+-------+----------+---------------+
| product_id | product_name     | category    | price | quantity | supplier      |
+------------+------------------+-------------+-------+----------+---------------+
|        101 | Laptop           | Electronics | 55000 |       15 | Dell          |
|        102 | Mouse            | Electronics |   799 |      120 | Logitech      |
|        103 | Keyboard         | Electronics |  1499 |       80 | HP            |
|        104 | Monitor          | Electronics | 12500 |       30 | Samsung       |
|        105 | Printer          | Electronics |  8500 |       20 | Canon         |
|        106 | Office Chair     | Furniture   |  4500 |       25 | Godrej        |
|        107 | Study Table      | Furniture   |  6500 |       18 | Nikamal       |
|        108 | Bookshelf        | Furniture   |  7200 |       12 | Godrej        |
|        109 | Sofa             | Furniture   | 28000 |        8 | Urban Ladder  |
|        110 | Dining Table     | Furniture   | 22000 |       10 | IKEA          |
|        111 | Water Bottle     | Kitchen     |   450 |      150 | Milton        |
|        112 | Mixer Grinder    | Kitchen     |  3800 |       22 | Prestige      |
|        113 | Pressure Cooker  | Kitchen     |  2800 |        8 | Hawkins       |
|        114 | Gas Stove        | Kitchen     |  5200 |       16 | Sunflame      |
|        115 | Microwave Oven   | Kitchen     |  9800 |       14 | LG            |
|        116 | Refrigerator     | Electronics | 36000 |        9 | Whirlpool     |
|        117 | Ceiling Fan      | Electronics |  2500 |       40 | Crompton      |
|        118 | Air conditioner  | Electronics | 42000 |        7 | C             |
|        119 | Smart TV         | Electronics | 48000 |       11 | Sony          |
|        120 | Iron Box         | Electronics |  1800 |       55 | Philips       |
+------------+------------------+-------------+-------+----------+---------------+
20 rows in set (0.00 sec)
```

4. Display only product names and prices.

```
mysql> select product_name,price from Product;
+------------------+-------+
| product_name     | price |
+------------------+-------+
| Laptop           | 55000 |
| Mouse            |   799 |
| Keyboard         |  1499 |
| Monitor          | 12500 |
| Printer          |  8500 |
| Office Chair     |  4500 |
| Study Table      |  6500 |
| Bookshelf        |  7200 |
| Sofa             | 28000 |
| Dining Table     | 22000 |
| Water Bottle     |   450 |
| Mixer Grinder    |  3800 |
| Pressure Cooker  |  2800 |
| Gas Stove        |  5200 |
| Microwave Oven   |  9800 |
| Refrigerator     | 36000 |
| Ceiling Fan      |  2500 |
| Air conditioner  | 42000 |
| Smart TV         | 48000 |
| Iron Box         |  1800 |
+------------------+-------+
20 rows in set (0.00 sec)
```

5. Find all Electronics products.

```
mysql> select *from Product where category = 'Electronics';
+------------+------------------+-------------+-------+----------+-----------+
| product_id | product_name     | category    | price | quantity | supplier  |
+------------+------------------+-------------+-------+----------+-----------+
|        101 | Laptop           | Electronics | 55000 |       15 | Dell      |
|        102 | Mouse            | Electronics |   799 |      120 | Logitech  |
|        103 | Keyboard         | Electronics |  1499 |       80 | HP        |
|        104 | Monitor          | Electronics | 12500 |       30 | Samsung   |
|        105 | Printer          | Electronics |  8500 |       20 | Canon     |
|        116 | Refrigerator     | Electronics | 36000 |        9 | Whirlpool |
|        117 | Ceiling Fan      | Electronics |  2500 |       40 | Crompton  |
|        118 | Air conditioner  | Electronics | 42000 |        7 | C         |
|        119 | Smart TV         | Electronics | 48000 |       11 | Sony      |
|        120 | Iron Box         | Electronics |  1800 |       55 | Philips   |
+------------+------------------+-------------+-------+----------+-----------+
10 rows in set (0.00 sec)
```

6. Display products whose price is greater than ₹10,000.

```
mysql> select * from Product where price > 10000;
+------------+------------------+-------------+-------+----------+---------------+
| product_id | product_name     | category    | price | quantity | supplier      |
+------------+------------------+-------------+-------+----------+---------------+
|        101 | Laptop           | Electronics | 55000 |       15 | Dell          |
|        104 | Monitor          | Electronics | 12500 |       30 | Samsung       |
|        109 | Sofa             | Furniture   | 28000 |        8 | Urban Ladder  |
|        110 | Dining Table     | Furniture   | 22000 |       10 | IKEA          |
|        116 | Refrigerator     | Electronics | 36000 |        9 | Whirlpool     |
|        118 | Air conditioner  | Electronics | 42000 |        7 | C             |
|        119 | Smart TV         | Electronics | 48000 |       11 | Sony          |
+------------+------------------+-------------+-------+----------+---------------+
7 rows in set (0.00 sec)
```

7. Find products whose quantity is less than 20.

```
mysql> select * from Product where quantity < 20;
+------------+------------------+-------------+-------+----------+---------------+
| product_id | product_name     | category    | price | quantity | supplier      |
+------------+------------------+-------------+-------+----------+---------------+
|        101 | Laptop           | Electronics | 55000 |       15 | Dell          |
|        107 | Study Table      | Furniture   |  6500 |       18 | Nikamal       |
|        108 | Bookshelf        | Furniture   |  7200 |       12 | Godrej        |
|        109 | Sofa             | Furniture   | 28000 |        8 | Urban Ladder  |
|        110 | Dining Table     | Furniture   | 22000 |       10 | IKEA          |
|        113 | Pressure Cooker  | Kitchen     |  2800 |        8 | Hawkins       |
|        114 | Gas Stove        | Kitchen     |  5200 |       16 | Sunflame      |
|        115 | Microwave Oven   | Kitchen     |  9800 |       14 | LG            |
|        116 | Refrigerator     | Electronics | 36000 |        9 | Whirlpool     |
|        118 | Air conditioner  | Electronics | 42000 |        7 | C             |
|        119 | Smart TV         | Electronics | 48000 |       11 | Sony          |
+------------+------------------+-------------+-------+----------+---------------+
11 rows in set (0.00 sec)
```

## Question 4: Library Management

### Table Structure

| Column | Data Type |
|---|---|
| book_id | INT (Primary Key) |
| title | VARCHAR(60) |
| author | VARCHAR(40) |
| category | VARCHAR(30) |
| price | INT |
| available | VARCHAR(5) |

**Sample Data (20 Records)**

| Book ID | Title | Author | Category | Price (₹) | Available |
|---|---|---|---|---|---|
| 201 | Java Programming | James Gosling | Programming | 650.00 | Yes |
| 202 | Python Basics | Guido van Rossum | Programming | 550.00 | Yes |
| 203 | SQL for Beginners | John Smith | Database | 480.00 | No |
| 204 | HTML & CSS | David Miller | Web Development | 420.00 | Yes |
| 205 | JavaScript Essentials | Brendan Eich | Web Development | 580.00 | Yes |
| 206 | React in Action | Mark Tielens Thomas | Web Development | 720.00 | No |
| 207 | Node.js Complete Guide | Ryan Dahl | Web Development | 690.00 | Yes |
| 208 | Database Concepts | Abraham Silberschatz | Database | 850.00 | Yes |
| 209 | Operating Systems | Andrew Tanenbaum | Computer Science | 780.00 | Yes |
| 210 | Computer Networks | James Kurose | Computer Science | 890.00 | No |
| 211 | Artificial Intelligence | Stuart Russell | AI | 950.00 | Yes |
| 212 | Machine Learning | Tom Mitchell | AI | 920.00 | Yes |
| 213 | Data Structures | Mark Allen Weiss | Programming | 610.00 | No |
| 214 | C Programming | Dennis Ritchie | Programming | 450.00 | Yes |
| 215 | Spring Boot Guide | Craig Walls | Programming | 810.00 | Yes |
| 216 | MongoDB Basics | Kristina Chodorow | Database | 540.00 | Yes |
| 217 | Docker Essentials | Nigel Poulton | Cloud Computing | 760.00 | No |
| 218 | AWS Cloud Practitioner | John Davis | Cloud Computing | 980.00 | Yes |
| 219 | Git & GitHub | Chris Dawson | Tools | 500.00 | Yes |
| 220 | Kubernetes Up & Running | Kelsey Hightower | Cloud Computing | 1100.00 | No |

**Practice Tasks**

1. Create above table

```
mysql> desc Books;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| book_id   | int         | NO   | PRI | NULL    |       |
| title     | varchar(60) | YES  |     | NULL    |       |
| author    | varchar(40) | YES  |     | NULL    |       |
| category  | varchar(30) | YES  |     | NULL    |       |
| price     | int         | YES  |     | NULL    |       |
| available | varchar(5)  | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

2. Insert all 20 records.

3. Display all books.

```
mysql> select * from Books;
+---------+-------------------------+---------------------+-------------------+-------+-----------+
| book_id | title                   | author              | category          | price | available |
+---------+-------------------------+---------------------+-------------------+-------+-----------+
|     201 | Java programming        | james goslinf       | programming       |   650 | yes       |
|     202 | Python Basics           | guido van rossum    | programming       |   550 | yes       |
|     203 | SQL for beginners       | john smith          | database          |   480 | no        |
|     204 | HTML & CSS              | david miller        | web development   |   420 | yes       |
|     205 | JavaScript Essentials   | Brendan Eich        | web development   |   580 | yes       |
|     206 | React in action         | Mark Tielens Thomas | web development   |   720 | no        |
|     207 | Node.js complete Guide  | Ryan Dahi           | web development   |   690 | yes       |
|     208 | Database Concepts       | Abraham Silberschatz| Database          |   850 | yes       |
|     209 | Operating Systems       | Andrew Tanenbaum    | Computer Science  |   780 | yes       |
|     210 | Computer Networks       | james kurose        | Computer Science  |   890 | no        |
|     211 | Artificial Inteligence  | stuart Rusell       | AI                |   950 | yes       |
|     212 | Machine Learning        | tom mitchell        | AI                |   920 | yes       |
|     213 | Data structures         | Mark Allen Weiss    | Prigramming       |   610 | no        |
|     214 | C Programming           | dennis ritchie      | Prigramming       |   540 | yes       |
|     215 | Spring boot guide       | craig walls         | Prigramming       |   810 | yes       |
|     216 | MongoDB basics          | kristina chodorow   | Database          |   540 | yes       |
|     217 | Docker Essentials       | Nigel Poutton       | Cloud Computing   |   760 | no        |
|     218 | AWS cloud Practitioner  | john davis          | cloud Computing   |   980 | yes       |
|     219 | Git & Github            | cris Dawson         | tools             |   500 | yes       |
|     220 | kubernetes up & running | kelsey hightower    | cloud computing   |  1100 | no        |
+---------+-------------------------+---------------------+-------------------+-------+-----------+
20 rows in set (0.00 sec)
```

4. Display only Title and Author

```
mysql> select title,author from Books;
+-------------------------+----------------------+
| title                   | author               |
+-------------------------+----------------------+
| Java programming        | james goslinf        |
| Python Basics           | guido van rossum     |
| SQL for beginners       | john smith           |
| HTML & CSS              | david miller         |
| JavaScript Essentials   | Brendan Eich         |
| React in action         | Mark Tielens Thomas  |
| Node.js complete Guide  | Ryan Dahi            |
| Database Concepts       | Abraham Silberschatz |
| Operating Systems       | Andrew Tanenbaum     |
| Computer Networks       | james kurose         |
| Artificial Inteligence  | stuart Rusell        |
| Machine Learning        | tom mitchell         |
| Data structures         | Mark Allen Weiss     |
| C Programming           | dennis ritchie       |
| Spring boot guide       | craig walls          |
| MongoDB basics          | kristina chodorow    |
| Docker Essentials       | Nigel Poutton        |
| AWS cloud Practitioner  | john davis           |
| Git & Github            | cris Dawson          |
| kubernetes up & running | kelsey hightower     |
+-------------------------+----------------------+
20 rows in set (0.00 sec)
```

.

5. Find all available books.

```
mysql> select * from Books where available = 'yes';
+---------+------------------------+----------------------+------------------+-------+-----------+
| book_id | title                  | author               | category         | price | available |
+---------+------------------------+----------------------+------------------+-------+-----------+
|     201 | Java programming       | james goslinf        | programming      |   650 | yes       |
|     202 | Python Basics          | guido van rossum     | programming      |   550 | yes       |
|     204 | HTML & CSS             | david miller         | web development  |   420 | yes       |
|     205 | JavaScript Essentials  | Brendan Eich         | web development  |   580 | yes       |
|     207 | Node.js complete Guide | Ryan Dahi            | web development  |   690 | yes       |
|     208 | Database Concepts      | Abraham Silberschatz | Database         |   850 | yes       |
|     209 | Operating Systems      | Andrew Tanenbaum     | Computer Science |   780 | yes       |
|     211 | Artificial Inteligence | stuart Rusell        | AI               |   950 | yes       |
|     212 | Machine Learning       | tom mitchell         | AI               |   920 | yes       |
|     214 | C Programming          | dennis ritchie       | Prigramming      |   540 | yes       |
|     215 | Spring boot guide      | craig walls          | Prigramming      |   810 | yes       |
|     216 | MongoDB basics         | kristina chodorow    | Database         |   540 | yes       |
|     218 | AWS cloud Practitioner | john davis           | cloud Computing  |   980 | yes       |
|     219 | Git & Github           | cris Dawson          | tools            |   500 | yes       |
+---------+------------------------+----------------------+------------------+-------+-----------+
14 rows in set (0.00 sec)
```

5. Display books whose price is greater than ₹700.

```
mysql> select * from Books where price > 700;
+---------+-------------------------+----------------------+------------------+-------+-----------+
| book_id | title                   | author               | category         | price | available |
+---------+-------------------------+----------------------+------------------+-------+-----------+
|     206 | React in action         | Mark Tielens Thomas  | web development  |   720 | no        |
|     208 | Database Concepts       | Abraham Silberschatz | Database         |   850 | yes       |
|     209 | Operating Systems       | Andrew Tanenbaum     | Computer Science |   780 | yes       |
|     210 | Computer Networks       | james kurose         | Computer Science |   890 | no        |
|     211 | Artificial Inteligence  | stuart Rusell        | AI               |   950 | yes       |
|     212 | Machine Learning        | tom mitchell         | AI               |   920 | yes       |
|     215 | Spring boot guide       | craig walls          | Prigramming      |   810 | yes       |
|     217 | Docker Essentials       | Nigel Poutton        | Cloud Computing  |   760 | no        |
|     218 | AWS cloud Practitioner  | john davis           | cloud Computing  |   980 | yes       |
|     220 | kubernetes up & running | kelsey hightower     | cloud computing  |  1100 | no        |
+---------+-------------------------+----------------------+------------------+-------+-----------+
10 rows in set (0.00 sec)
```

6. Find all books in the Programming category.

```
mysql> select * from Books where category = 'programming';
+---------+--------------------+-------------------+-------------+-------+-----------+
| book_id | title              | author            | category    | price | available |
+---------+--------------------+-------------------+-------------+-------+-----------+
|     201 | Java programming   | james goslinf     | programming |   650 | yes       |
|     202 | Python Basics      | guido van rossum  | programming |   550 | yes       |
+---------+--------------------+-------------------+-------------+-------+-----------+
2 rows in set (0.00 sec)
```

### Question 5: Customer Orders

### Table Structure

| Column | Data Type |
|---|---|
| order_id | INT (Primary Key) |
| customer_name | VARCHAR(50) |
| product | VARCHAR(50) |
| quantity | INT |
| amount | INT |
| city | VARCHAR(30) |

### Sample Data (20 Records)

| Order ID | Customer Name | Product Name | Category | Quantity | Amount (₹) | Order Date | City |
|---|---|---|---|---|---|---|---|
| 301 | Amit Sharma | Laptop | Electronics | 1 | 55000.00 | 2026-07-01 | Delhi |
| 302 | Priya Verma | Mouse | Electronics | 2 | 1600.00 | 2026-07-02 | Pune |
| 303 | Rahul Singh | Keyboard | Electronics | 1 | 1500.00 | 2026-07-02 | Mumbai |
| 304 | Sneha Patel | Smart TV | Electronics | 1 | 48000.00 | 2026-07-03 | Delhi |
| 305 | Rohan Gupta | Office Chair | Furniture | 2 | 9000.00 | 2026-07-03 | Jaipur |
| 306 | Anjali Mehta | Refrigerator | Electronics | 1 | 36000.00 | 2026-07-04 | Pune |
| 307 | Vikas Kumar | Microwave Oven | Kitchen | 1 | 9800.00 | 2026-07-04 | Mumbai |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 308 | Neha Joshi | Water Bottle | Kitchen | 4 | 1800.00 | 2026-07-05 | Delhi |
| 309 | Arjun Mishra | Study Table | Furniture | 1 | 6500.00 | 2026-07-05 | Lucknow |
| 310 | Kiran Yadav | Air Conditioner | Electronics | 1 | 42000.00 | 2026-07-06 | Pune |
| 311 | Meera Shah | Ceiling Fan | Electronics | 2 | 5000.00 | 2026-07-06 | Delhi |
| 312 | Nikhil Jain | Mixer Grinder | Kitchen | 1 | 3800.00 | 2026-07-07 | Mumbai |
| 313 | Pooja Roy | Pressure Cooker | Kitchen | 2 | 5600.00 | 2026-07-07 | Jaipur |
| 314 | Ritesh Das | Printer | Electronics | 1 | 8500.00 | 2026-07-08 | Pune |
| 315 | Komal Kapoor | Sofa | Furniture | 1 | 28000.00 | 2026-07-08 | Delhi |
| 316 | Deepak Singh | Iron Box | Electronics | 3 | 5400.00 | 2026-07-09 | Mumbai |
| 317 | Sonal Agarwal | Laptop | Electronics | 1 | 55000.00 | 2026-07-09 | Pune |
| 318 | Manish Verma | Bookshelf | Furniture | 1 | 7200.00 | 2026-07-10 | Delhi |
| 319 | Ankit Tiwari | Monitor | Electronics | 2 | 25000.00 | 2026-07-10 | Lucknow |
| 320 | Payal Sharma | Gas Stove | Kitchen | 1 | 5200.00 | 2026-07-11 | Jaipur |

**Practice Tasks**

1. Create above table.

```
mysql> desc Customer_order;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| order_id      | int         | NO   | PRI | NULL    |       |
| customer_name | varchar(50) | YES  |     | NULL    |       |
| product       | varchar(50) | YES  |     | NULL    |       |
| category      | varchar(40) | YES  |     | NULL    |       |
| quantity      | int         | YES  |     | NULL    |       |
| amount        | int         | YES  |     | NULL    |       |
| order_date    | varchar(50) | YES  |     | NULL    |       |
| city          | varchar(30) | YES  |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

2. Insert all 20 records.

3. Display all orders.

```
mysql> select * from Customer_order;
+----------+---------------+------------------+-------------+----------+--------+------------+---------+
| order_id | customer_name | product          | category    | quantity | amount | order_date | city    |
+----------+---------------+------------------+-------------+----------+--------+------------+---------+
|      301 | Amit Sharma   | Laptop           | Electronics |        1 |  55000 | 2026-07-01 | Delhi   |
|      302 | Priya Verma   | Mouse            | Electronics |        2 |   1600 | 2026-07-02 | Pune    |
|      303 | Rahul Singh   | Keyboard         | Electronics |        1 |   1500 | 2026-07-02 | Mumbai  |
|      304 | Sneha Patel   | Smart TV         | Electronics |        1 |  48000 | 2026-07-03 | Delhi   |
|      305 | Rohan Gupta   | Office chair     | Furniture   |        2 |   9000 | 2026-07-03 | Jaipur  |
|      306 | Anjali Mehta  | Refrigerator     | Electronics |        1 |  36000 | 2026-07-04 | Pune    |
|      307 | Vikas Kumar   | Microwave Oven   | Kitchen     |        1 |   9800 | 2026-07-04 | Mumbai  |
|      308 | Neha Joshi    | Water bottle     | Kitchen     |        4 |   1800 | 2026-07-04 | Mumbai  |
|      309 | Arjun Mishra  | Study Table      | Furniture   |        1 |   6500 | 2026-07-05 | Lucknow |
|      310 | Kiran Yadav   | Air conditioner  | Electronics |        1 |  42000 | 2026-07-06 | Pune    |
|      311 | Meera Shah    | Ceiling Fan      | Electronics |        2 |   5000 | 2026-07-06 | Delhi   |
|      312 | Nikhil Jain   | MIxer Grinder    | Kitchen     |        1 |   3800 | 2026-07-07 | Mumbai  |
|      313 | Pooja Roy     | Pressure Cooker  | Kitchen     |        2 |   5600 | 2026-07-07 | Jaipur  |
|      314 | Ritesh Das    | Printer          | Electronics |        1 |   8500 | 2026-07-08 | Pune    |
|      315 | Komal Das     | Sofa             | Furniture   |        1 |  28000 | 2026-07-08 | Delhi   |
|      316 | Deepak Singh  | Iron Box         | Electronics |        3 |   5400 | 2026-07-09 | Mumbai  |
|      317 | Sonal Agarwal | Laptop           | Electronics |        1 |  55000 | 2026-07-09 | Pune    |
|      318 | Manish Verma  | Bookshelf        | Furniture   |        1 |   7200 | 2026-07-10 | Delhi   |
|      319 | Ankit Tiwari  | Monitor          | Electronics |        2 |  25000 | 2026-07-10 | Lucknow |
|      320 | Payal Sharma  | gas stove        | Kitchen     |        1 |   5200 | 2026-07-11 | Jaipur  |
+----------+---------------+------------------+-------------+----------+--------+------------+---------+
20 rows in set (0.00 sec)
```

4. Display customer names and products.

```
mysql> select customer_name,product from Customer_order;
+---------------+------------------+
| customer_name | product          |
+---------------+------------------+
| Amit Sharma   | Laptop           |
| Priya Verma   | Mouse            |
| Rahul Singh   | Keyboard         |
| Sneha Patel   | Smart TV         |
| Rohan Gupta   | Office chair     |
| Anjali Mehta  | Refrigerator     |
| Vikas Kumar   | Microwave Oven   |
| Neha Joshi    | Water bottle     |
| Arjun Mishra  | Study Table      |
| Kiran Yadav   | Air conditioner  |
| Meera Shah    | Ceiling Fan      |
| Nikhil Jain   | MIxer Grinder    |
| Pooja Roy     | Pressure Cooker  |
| Ritesh Das    | Printer          |
| Komal Das     | Sofa             |
| Deepak Singh  | Iron Box         |
| Sonal Agarwal | Laptop           |
| Manish Verma  | Bookshelf        |
| Ankit Tiwari  | Monitor          |
| Payal Sharma  | gas stove        |
+---------------+------------------+
20 rows in set (0.00 sec)
```

5. Find all orders placed from Pune.
6. Display orders where the amount is greater than ₹20,000.
7. Find orders where the quantity is greater than 2.

```
mysql> select * from Customer_order where city = 'Pune';
+----------+---------------+-----------------+-------------+----------+--------+------------+------+
| order_id | customer_name | product         | category    | quantity | amount | order_date | city |
+----------+---------------+-----------------+-------------+----------+--------+------------+------+
|      302 | Priya Verma   | Mouse           | Electronics |        2 |   1600 | 2026-07-02 | Pune |
|      306 | Anjali Mehta  | Refrigerator    | Electronics |        1 |  36000 | 2026-07-04 | Pune |
|      310 | Kiran Yadav   | Air conditioner | Electronics |        1 |  42000 | 2026-07-06 | Pune |
|      314 | Ritesh Das    | Printer         | Electronics |        1 |   8500 | 2026-07-08 | Pune |
|      317 | Sonal Agarwal | Laptop          | Electronics |        1 |  55000 | 2026-07-09 | Pune |
+----------+---------------+-----------------+-------------+----------+--------+------------+------+
5 rows in set (0.00 sec)

mysql> select * from Customer_order where amount > 20000;
+----------+---------------+-----------------+-------------+----------+--------+------------+---------+
| order_id | customer_name | product         | category    | quantity | amount | order_date | city    |
+----------+---------------+-----------------+-------------+----------+--------+------------+---------+
|      301 | Amit Sharma   | Laptop          | Electronics |        1 |  55000 | 2026-07-01 | Delhi   |
|      304 | Sneha Patel   | Smart TV        | Electronics |        1 |  48000 | 2026-07-03 | Delhi   |
|      306 | Anjali Mehta  | Refrigerator    | Electronics |        1 |  36000 | 2026-07-04 | Pune    |
|      310 | Kiran Yadav   | Air conditioner | Electronics |        1 |  42000 | 2026-07-06 | Pune    |
|      315 | Komal Das     | Sofa            | Furniture   |        1 |  28000 | 2026-07-08 | Delhi   |
|      317 | Sonal Agarwal | Laptop          | Electronics |        1 |  55000 | 2026-07-09 | Pune    |
|      319 | Ankit Tiwari  | Monitor         | Electronics |        2 |  25000 | 2026-07-10 | Lucknow |
+----------+---------------+-----------------+-------------+----------+--------+------------+---------+
7 rows in set (0.00 sec)

mysql> select * from Customer_order where quantity > 2;
+----------+---------------+--------------+-------------+----------+--------+------------+--------+
| order_id | customer_name | product      | category    | quantity | amount | order_date | city   |
+----------+---------------+--------------+-------------+----------+--------+------------+--------+
|      308 | Neha Joshi    | Water bottle | Kitchen     |        4 |   1800 | 2026-07-04 | Mumbai |
|      316 | Deepak Singh  | Iron Box     | Electronics |        3 |   5400 | 2026-07-09 | Mumbai |
+----------+---------------+--------------+-------------+----------+--------+------------+--------+
2 rows in set (0.00 sec)
```